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Point of View

Concerns on the Science and Practice of a Movement System

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Introduction

Theoretically, the movement system is the integration of body systems that generate and maintain all movements contributing to individual function.[1] Movement diagnoses are diagnostic labels ascribed to movements that are hypothesized to be aberrant.[2] In the context of the International Classification of Functioning, Disability and Health framework, a movement diagnosis may be most appropriately positioned as a "body systems impairment." Recently, movement diagnoses have been put forth as fundamental to the movement system that is being advocated as a model for physical therapist management.[3] However, we caution against expeditious advancement of practice approaches that have yet to be empirically substantiated. In this Point of View, we look at the science behind a movement system diagnosis. We highlight scientific challenges facing the adoption of the proposed movement system as a cornerstone of clinical practice as it relates to the International Classification of Functioning, Disability, and Health model.

Key Clinical and Psychometric Properties of a Movement Diagnosis

The Balance Diagnosis Task Force within the Academy of Neurological Physical Therapy recently published a classification system for balance dysfunctions that included 10 proposed movement diagnoses.[4] These diagnoses are considered important for detecting underlying causes of dysfunctional balance and informing intervention selection. We use 1 of these diagnoses, "deficits in steady state postural control – verticality" (termed "verticality") to exemplify an empirical process that would support the clinical utility of this diagnostic label.

The presence of the movement diagnosis should be relevant to a health condition or pathology, an activity limitation, or a participation restriction. For example, verticality should be associated with an activity limitation such as falling. To determine this, patients with and without a history of falling would be assessed for verticality by 2 clinicians blinded to the patients' history of falls. This research, considered known-group validation,[5] would inform whether verticality is just a benign finding versus a movement impairment associated with an activity limitation. To further investigate clinical relevance of verticality from a prognostic perspective, an assessment for verticality in participants could be completed before there are any documented falls. Participants would then be longitudinally tracked to determine fall status in those with and without verticality. If a verticality diagnosis does not predict future falls, then it lacks predictive validity and can be considered a normal movement variant. Establishing known-group and predictive validity is an integral step in the development of a movement diagnosis. If the presence of a movement diagnosis is not associated with a health condition, activity limitation, or participation restriction, then it may be unwarranted.

Ideally, and particularly relevant to physical therapy practitioners, the presence of a movement diagnosis should also inform treatment decision making. Continuing our example, researchers can sample a group of patients who have a history of falls, measure their verticality at baseline, and then compare 2 different types of treatment (treatment "A" and "B") and subsequent fall outcomes. If the presence of verticality impacts the response to treatment (ie, those with better verticality experience a decrease in falls after the treatment A compared to treatment B), then verticality would be considered an important treatment effect modifier. In other words, identifying patients with verticality would influence the selection of a specific intervention (in this case intervention "A") to produce a favorable outcome. On the other hand, if the magnitude of verticality at evaluation had no bearing on a favorable treatment outcome, then it becomes less important to identify patients with verticality to inform treatment decision making.

Last, treatment of a movement diagnosis should have influence along the causal pathway of an outcome. This newer concept in rehabilitation research extends common comparative outcome design and requires specific measurement and analysis. A group of patients with a history of falling can be randomly split, with 1 group receiving an intervention to specifically address verticality and the other group receiving a general standard of care treatment. Then, 2 separate analyses can be performed. First, a traditional comparative outcomes analysis will determine if the intervention was more effective for reducing falls compared to the general intervention. Second, a causal mediation analysis can determine

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whether a change in verticality was different between the 2 groups and preceded the improvement in falls outcome. In other words, did the intervention cause a change in verticality compared to the standard of care intervention, and did that change in verticality cause the improvement in falls. The second analysis determines if verticality is a mechanism on the causal pathway of an outcome (ie, falls in this example). In this scenario, verticality would be considered a treatment mediator. In other words, changing verticality mediated an improvement in falls and, therefore, verticality is a movement impairment that should be targeted with treatment.

It is important to note here that an improvement in clinical outcomes, such as decreasing falls, does not provide support for the movement diagnosis. In our above scenario, if the first comparative outcomes analysis revealed a larger improvement in the verticality treatment, this is interpreted as a better treatment than the standard of care for patients with fall risks. However, concluding that the improvement was due to the classification of a verticality movement diagnosis would be a post-hoc fallacy. Moreover, assuming that the larger effect in the verticality intervention was caused by a change in verticality is false. Improvements in fall risk can be attributed to a multitude of mechanisms unrelated to a movement diagnosis such as unequal comparator interventions, placebo and nonspecific contextual effects, regression to the mean, changes in social determinants of falls, or other peripheral and central neurophysiological changes. Randomized trial results offer guidance on which treatments are effective, but unless the comparative outcome analysis is accompanied with a mediator analysis, they do not offer guidance on why the treatment was effective or if a movement diagnosis was warranted. One other important note is that we have built this example under the assumption that verticality is in fact an actual construct and clinicians can reliably detect verticality. However, this is not a safe assumption particularly in movements that are difficult to discern. A scientifically and clinically sound diagnostic label takes rigorous and methodical investigation, as described above, that should precede widespread implementation.

Existing Literature Does Not Support Foundational Aspects of Movement System Diagnoses

Empirical issues exist for patients with musculoskeletal conditions. Recently, the American Physical Therapy Association created a Movement System Diagnosis Template to solicit ideas for movement diagnoses and build the movement system. They used an example of proposed movement diagnosis to help guide contributors.[6] The exemplar dysfunction was scapular upward rotation deficit (SURD); however, the scientific support for SURD is tenuous. We briefly present some of this evidence below:

- In terms of reliability, a 2017 systematic review reported wide diversity in performance, description, and interpretation of tests along with low-quality study methodology and inadequate between-rater reliability.[7]
- In terms of known-groups validity, Plummer et al[8] and Uhl et al[9] found no significant difference in shoulder pain between those with and without scapula dyskinesis.
- In terms of predictive validity, some studies suggest that scapular dyskinesis predicts injury risk,[10,11] while others disagree.[12,13]
- In terms of treatment relevance, Turgut et al[14] found no difference in 12-week shoulder pain and disability between a scapular motor control group versus general shoulder rehabilitation despite a reduction in dyskinesia in the motor control group. Similarly, in an 8-week observational cohort of patients with subacromial shoulder pain, change in shoulder upward rotation was not associated with improvement in pain.[15]
- In terms of treatment mechanisms, Hotta et al[16] randomized people with shoulder pain to receive scapular strengthening or periscapular strengthening over the course of 8 weeks. There was no difference in function or pain outcomes between the 2 groups. Additionally, they measured putative mediators such as change in scapular motion and muscle strength of scapulothoracic muscles midway through the study to see if changing these mediators is associated with clinical outcomes. There was no effect of any hypothesized mediators on pain or function.

Collectively, the existing literature investigating SURD gives us pause about confidently ascribing scapular dysfunction as a movement diagnosis that causes pain and should be an intervention target. Similar issues exist for knee valgus,[17,18] neck posture,[19] and patella tracking.[20,21] This is concerning considering the implications of building a diagnostic classification schema upon proposed movement dysfunctions with limited scientific support for psychometric validity and clinical application.

Moving Forward, Part 1: Shifting Away From Diagnostic Labels

Across the medical field, the importance of diagnostic labelling for many chronic non-communicable conditions (ie, those in which patients commonly seek physical therapy services) is being scrutinized, particularly when measured against the value of an accurate and meaningful prognosis.[22–24] In addition, emphasizing diagnoses has been associated with overmedicalization and overtreatment.[25,26] It has also been reported that environmental and personal factors (eg, socioeconomic status, psychological co-morbidities) are strongly associated with clinical outcomes.[27] We see relevance of all these concerns pertaining to the movement system. A movement diagnosis that is, in fact, based on a benign aberrant movement inherently may have limited prognostic value, can potentially lead to unnecessary intervention, and might detract the clinician from providing an accurate forecast of what are the most likely long-term outcomes for a given individual.

Moving Forward, Part 2: Better Appreciation of Movement Variability

Assuming an idealistic norm based on the study of a very narrow and homogenous population limits a full understanding of what constitutes "normal" and "abnormal" movement. In contrast to what is being proposed, "normal movement" may be difficult to determine because it is a product of interacting dynamic systems that accomplish a task within an environment.[28] Additionally, there is tremendous variability expected in movement based on different patient-specific factors: culture, race, and medical conditions all have the potential to

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influence movement variability. Therefore, normal movement is highly contextual and what constitutes normal movement should be expected to be different based on what is needed to accomplish a goal in a particular environment. Confining movement assessment into uniplanar segments under artificial environments typical in clinical settings (as has been historically done and is being proposed in current assessment schema) is not realistic to how movement operates in society. Instead, assessment could be focused on the ability of the person to accomplish various self-selected tasks, under different internal and external constraints, in an environment similar to their life. Then, treatment of a movement would not focus on normalizing presupposed impairments but rather creating real-world movement proficiency and adaptability.

Determination of what constitutes normal movement variability could be established using populations of different races, culture, pain status, and medical diagnosis. In addition, intervention studies that measure putative biological, psychological, and sociological treatment mechanisms would elucidate the salient drivers of clinical outcomes. And last, identifying prognostic indicators would guide the clinician toward targeting specific treatments based on patient-specific factors. Until then, we fear that a system prematurely constructs movement dysfunctions and purports them as a cause for pain and disability may perpetuate the growth of unfounded assessments and interventions and, in doing so, work counterproductively to establish our profession by who we are, not what we do.

Conclusion

We implore a cautious, methodical, and scientifically sound approach moving forward that takes into consideration some key aspects in human movement, research methodology, and prognostic medicine. The issues we mention in this Point of View are not unique to the proposed movement system; indeed, how the inaccuracy of diagnostic labeling limits patient management is a recurring theme in clinical sciences. However, the proposed movement system provides the appropriate context for this Point of View, as it is widely endorsed as a framework for future physical therapist practice, despite using diagnostic labels that lack scientific support. Thus, a focused look at the implications of diagnostic labeling with direct relevance to the movement system was warranted.

Author Contributions

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Writing: C. Joyce, J. Beneciuk, S.Z. George
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